Structural Foundations for Abstract Mathematics

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What no foundation can give us:

- Certainty

What foundations need not give us:

- Ontological reduction

What set theory gives us:

- Common language in which *all* mathematics can be encoded: $\forall, \exists, \in, \ldots$
- Dispute resolution (is CH true?)
- Guidance in gray areas (can we assign a measure to all subsets of $\mathbb{R}$?)
- Clarity about inferential relations and commitments

When set theory throws sand in our eyes:

- The Doplicher-Roberts theorem
- Two theories cannot be equivalent unless their models are isomorphic
- Equivalence is *the* correct notion of "sameness" for categories.
- Benacerraf: number theory is insensitive to set-theoretic details
- $\mathbb{Z} \cong 2\mathbb{Z}$, but there is a set-theoretic predicate P such that $P(\mathbb{Z})$ and $\neg P(2\mathbb{Z})$.

$$P(x) =_{\mathrm{df}} 1 \in x$$

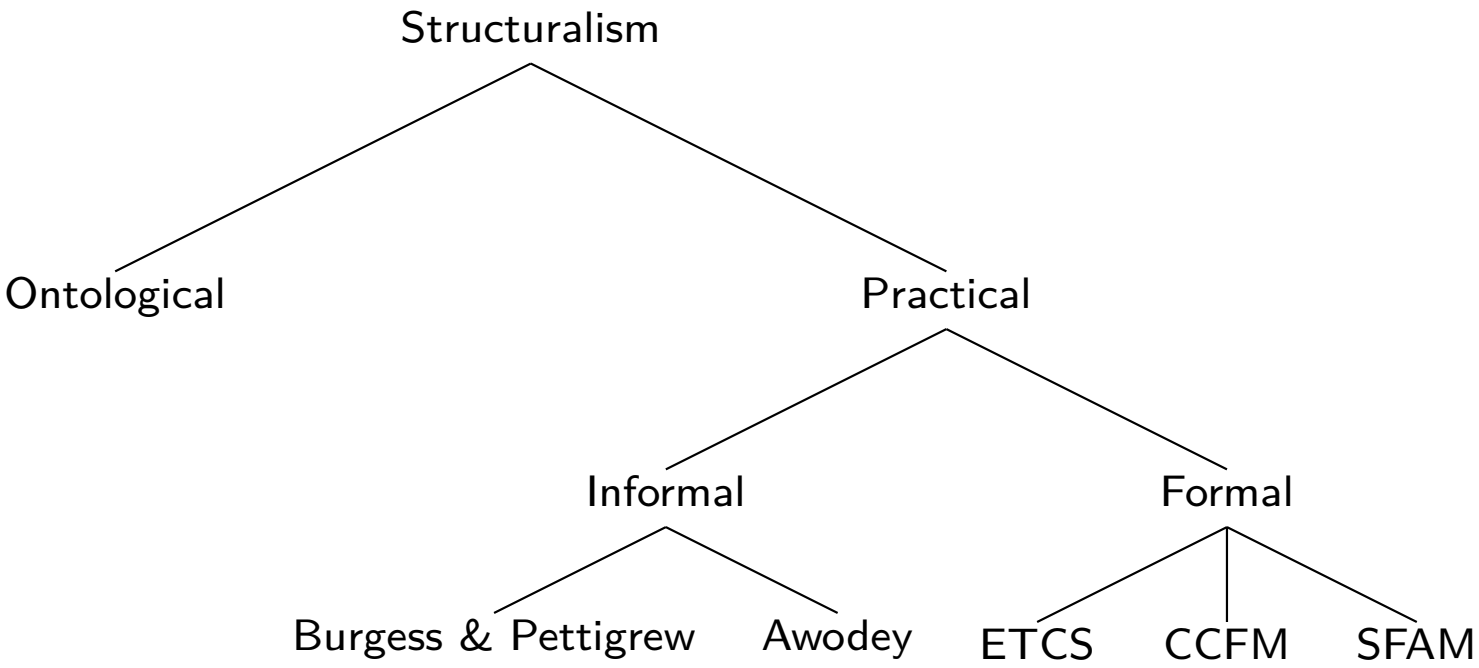
- Makkai:

  "Let G and H be arbitrary groups, and consider the intersection of their underlying sets."

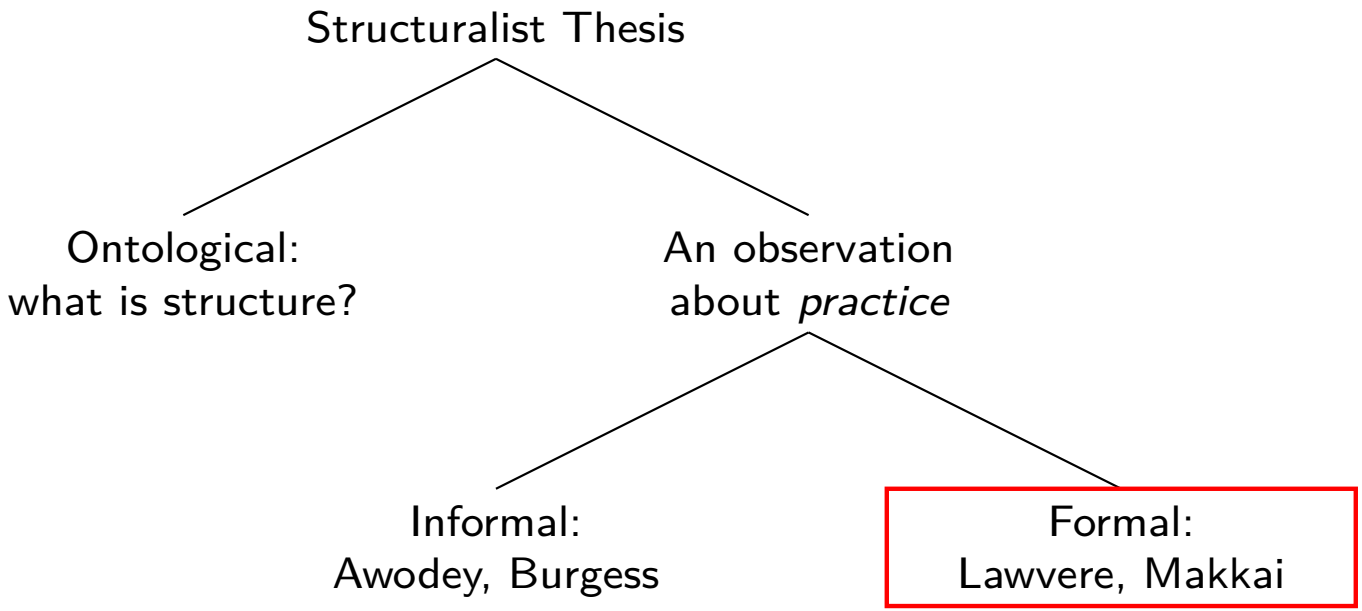
- MacLane:

  "... as Weyl once remarked, [set theory] contains far too much sand."

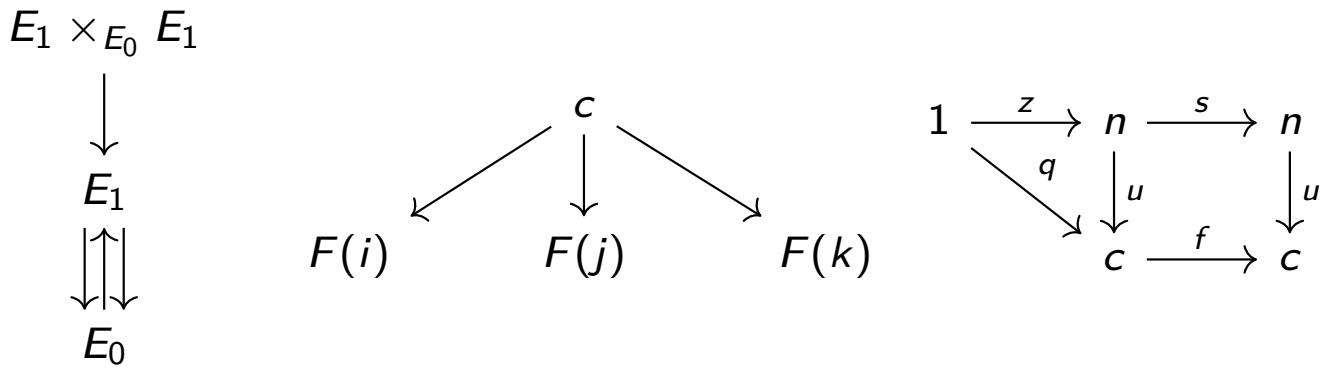
Proposed solutions to the **sand problem**:



Structuralist Thesis: Mathematics is concerned with the relations that objects bear to each other, rather than with what these objects *are*.



Elementary Theory of the Category of Sets

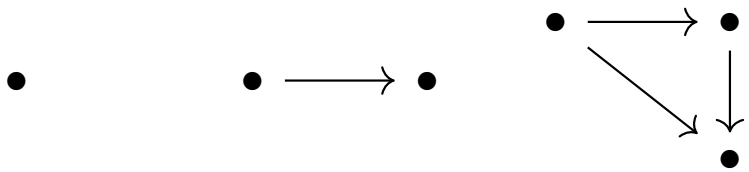


- Two-sorted theory
- No Sand: Quantifiers range over objects and arrows, not over elements
- But: $a \in S \quad \equiv \quad 1 \xrightarrow{a} S$

Category of Categories as the Foundation for Mathematics

Language: $\{\Delta_0, \Delta_1, \cdot\} \cup \{\partial_0, \partial_1\}$

1	**2**	**3**
$\forall y \exists! z (y \xrightarrow{z} x)$	$\Delta_i(\partial_j) = \mathbf{2}$	



"arrow in C" $\equiv$ "functor from $\mathbf{2}$ to A"

Global foundations

	Syntax	Language	Deductive System	Semantics
Set Theory	$\mathrm{FOL}_{=}$	$\{\in\}$	classical + ZFC	cumulative hierarchy
ETCS	$\mathrm{FOL}_{=}$	$\{\circ, d_0, d_1, i\}$	classical + Lawvere	CCFM?

In all cases, $=$ is globally defined

All global foundations say "stupid" things

Claim Any global foundation will say stupid things.

- Levels: A set is a structure of level 0. A mathematical structure is of level n (incl. ∞) if its natural setting is in an n-category.
  - e.g. Groups are level-1 structures, categories are level-2
  - e.g. Simplicial sets are both level 1 and level ∞ depending on the context
- Suppose we have two local criteria of identity $\sim_1$ and $\sim_2$. We say that $\sim_1$ is **coarser** (resp. **finer**) than $\sim_2$ if $P \sim_2 Q$ implies $P \sim_1 Q$ (resp. $P \sim_1 Q$ implies $P \sim_2 Q$)
  - This assumes that structures are commensurable, possible up to some canonical mapping. e.g. $\mathbb{Q} \hookrightarrow \mathbb{R}$.
- "Lemma": If $m \geq n$ then the criterion of identity $\sim_{\mathcal{S}}$ for a collection (or type) $\mathcal{S}$ of structures of level m will be coarser than the criterion of identity $\sim_{\mathcal{T}}$ for a collection (or type) $\mathcal{T}$ of structures of of level n.
  - "Proof" by examples: Group objects in **Set**, Category objects in a topos $\mathcal{E}$.

Permanent Parameter Structuralism

Proposal: Treat $\mathbb{R}$, $\mathbb{N}$, etc. as arbitrary names, i.e. they name an arbitrary one of the individuals satisfying certain properties

Problems:

- Informal: Doesn't clarify our inferential rules for arbitrary structures
- Doesn't have any predictive value

Local criteria of structural identity

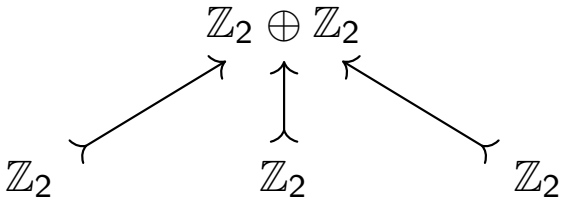
- Isomorphism of groups preserves all "group theoretic" concepts and properties
- Equivalence of categories preserves all "categorical" concepts and properties: e.g. having certain limits or colimits
- Homotopy equivalence of spaces

A mathematical practice determines a local notion of identity. Within this practice, if $a \cong b$ *then* $\models \phi(a) \leftrightarrow \phi(b)$ *for any well-formed formula* $\phi(x)$.

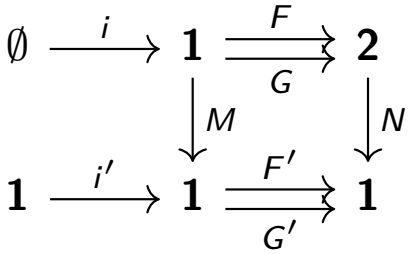
Against naive structuralism

But $\cong$ doesn't mean $=$

- Worse than stupid: $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ has one proper subgroup



- Limits in **Cat** are not invariant under categorical equivalences

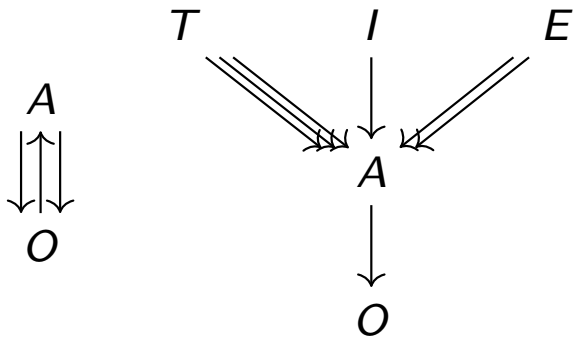


Makkai's Structural Foundations for Abstract Mathematics

	Syntax	Language	Deductive System	Semantics
SFAM	FOLDS	FOLDS signatures	classical	weak ∞-categories

FOLDS

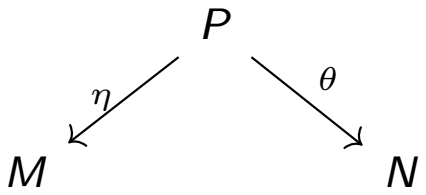
- FOLDS syntax: multi-sorted FOL with sort dependence
- FOLDS signatures: One-way, skeletal, simple categories



- FOLDS semantics: functors into the meta-category $\mathcal{S}$

FOLDS equivalence

Isomorphic Structures: Two $\mathcal{L}$-structures $M, N : \mathcal{L} \to \mathcal{S}$ are said to be **FOLDS equivalent** if there exists an $\mathcal{L}$-structure P and fiberwise surjective natural transformations η and θ giving a span of the form:



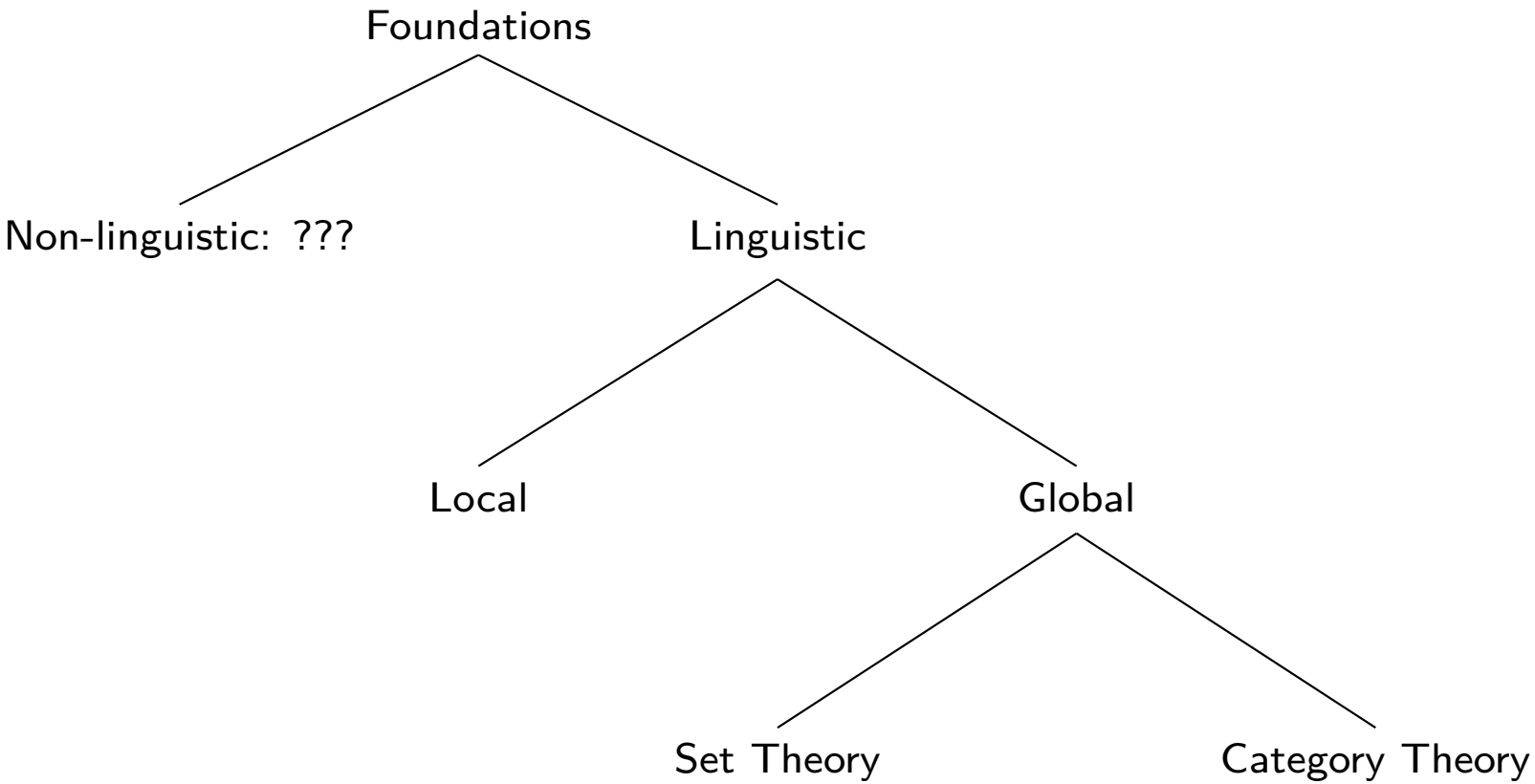
FOLDS results

- Indiscernibility of Isomorphs: If $M \models \phi$ and $M \cong_{\mathcal{L}} N$ then $N \models \phi$.
- Inductive evidence for correctness of FOLDS
  - Two level 1 mathematical structures M and N (e.g. groups, fields) are FOLDS equivalent just in case they are isomorphic.
  - Two categories M and N are folds equivalent just in case they are equivalent as categories.
- Makkai's Conjecture: For each n, there is a signature $\mathcal{L}_n$ corresponding to n-categories, and two $\mathcal{L}_n$ structures are FOLDS equivalent just in case that are Baez-Dolan equivalent as n-categories.
- Corollary: In FOLDS, it's impossible to be *evil* or *stupid*.

Problems for FOLDS

- FOLDS is either like first-order model theory, or like ZF set theory
- If the metatheory for FOLDS is formal, then it's global.

A more radical classification of foundations



General argument against linguistic foundations

- Global: ZFC, ETCS, CCAF. There is a fixed language and a fixed criterion of identity.
- Local: SFAM, Bell's local mathematics, model theory, Bourbaki style structuralism.

Foundations without language: MLTT, HoTT, UF

Four types of judgments

$\Gamma \vdash A\ \mathbf{Type}$	A is a type
$\Gamma \vdash A = B\ \mathbf{Type}$	A and B are the same type
$\Gamma \vdash a : A$	a is a term of type A
$\Gamma \vdash a = b : A$	a and b are the same term of type A

$$\frac{\Gamma \vdash A : \mathcal{U} \qquad \Gamma \vdash a : A \qquad \Gamma \vdash b : A}{\Gamma \vdash a =_A b : \mathcal{U}}\ \mathtt{Id}\text{-FORM} \qquad \frac{\Gamma \vdash A : \mathcal{U} \qquad \Gamma \vdash a : A}{\Gamma \vdash \mathtt{refl}_a : a =_A a}\ \mathtt{Id}\text{-INTRO}$$

$$\frac{\Gamma, x : A, y : A, p : x =_A y \vdash C : \mathcal{U} \qquad \Gamma, z : A \vdash d(z) : C(z, z, \mathtt{refl}_z)}{\Gamma \vdash D : \prod_{a,b,p'} C(a, b, p')}\ \mathtt{Id}\text{-ELIM}$$

$$\frac{\begin{array}{c}\Gamma, x : A, y : A, p : x =_A y \vdash C : \mathcal{U} \\ \Gamma, z : A \vdash d : C(z, z, \mathtt{refl}_z) \qquad \Gamma \vdash a : A\end{array}}{\Gamma \vdash D(a, a, \mathtt{refl}_a) = d(a) : C(a, a, \mathtt{refl}_a)}\ \mathtt{Id}\text{-COMP}$$

Type Theory	Logic	Set Theory	HoTT
A	proposition	set	space
$a : A$	proof	element	point
$B(x)$	predicate	family of sets	fibration
$b(x) : B(x)$	conditional proof	family of elements	section
$0, 1$	$\bot, \top$	$\emptyset, \{\emptyset\}$	$\emptyset, *$
$A + B$	$A \vee B$	disjoint union	coproduct
$A \times B$	$A \wedge B$	set of pairs	product space
$A \to B$	$A \Rightarrow B$	set of functions	function space
$\sum_{x:\,A} B(x)$	$\exists x \in A B(x)$	disjoint sum	total space
$\prod_{x:\,A} B(x)$	$\forall x \in A B(x)$	product	space of sections
$\mathsf{Id}_A(x, y)$	$x = y$	$\{\langle x, x\rangle \mid x \in A\}$	path space A^I

Table: Points of view of Type Theory

Univalence for Dummies

Overheard: "Isomorphism *is* Identity"

$$\mathtt{isEquiv}(f) = \prod_{x\colon A} \mathtt{isContr}(\mathtt{hFib}(f, x))$$

$$\mathtt{univ}\colon \prod_{A,B\colon \mathcal{U}} \mathtt{isEquiv}(\mathtt{idtoequiv})$$

where `idtoequiv` is the canonical map guaranteed by induction. Or in a more informal manner:

$$(A = B) \simeq (A \simeq B)$$

YES: Isomorphism is Isomorphic to Identity

Univalence as a transport principle

The Univalence Axiom should, at least from a logical/foundational point of view, be viewed as a "transport" principle: it allows transport of any proof about a "structure" (i.e. a type) to any "structure" that is equivalent to it, via the identity type.
Type-theoretic properties are invariant under types for which a proof of identity can be produced, and therefore properties will be invariant under equivalent types too.

Connecting FOLDS and UF

Using (Ahrendt, Kapulkin, Shulman 2013), we can show:

Theorem

Every expressible property of an object in a category is invariant under isomorphism.

Proof.

Take a category $\mathcal{C}$ with object type C. A property of an object in $\mathcal{C}$ is a type $C \to \mathbf{Prop}$. Suppose $C(a)$ holds for some $a : C$, i.e. there exists a term (proof) $p : C(a)$ and suppose also that there is an isomorphism between a and b, i.e. a term $\eta : a \cong b$. Now, by the condition, the canonical map $\mathtt{idtoiso}(a, b)$ has a quasi-inverse $\mathtt{isotoid}(a, b)$. Thus we get a term

$$\epsilon =_{\mathsf{df}} \mathtt{isotoid}(a, b)(\eta) : \mathtt{Id}_C(a, b)$$

Since we now have a proof of identity we may transfer any property that we can express of a along it and construct a proof that that property holds also of the other identificand. More precisely we use the `transport` function to transfer the proof p along ϵ, thus getting a term $\mathtt{transport}(\epsilon)(p) : C(b)$, as required. □

Connecting FOLDS and UF

Theorem (Makkai-Tsementzis) Let S, T be Kan complexes. Then S and T are homotopy equivalent if and only if $i^*S \simeq_{\Delta_+^{\mathrm{op}}} i^*T$ (i.e. if and only if i^*S and i^*T are FOLDS-equivalent as FOLDS Δ_+^{op}-structures.)